Packet Tracer Switch Security Configuration

Completed	☑
Status	To do: Lab Work

Introduction

The primary goal for this activity is to reinforce understanding and practical application of security configurations at the data link layer of the network. The lab utilizes standard Cisco networking equipment.

Objective

Part 1: Configure the Network Devices

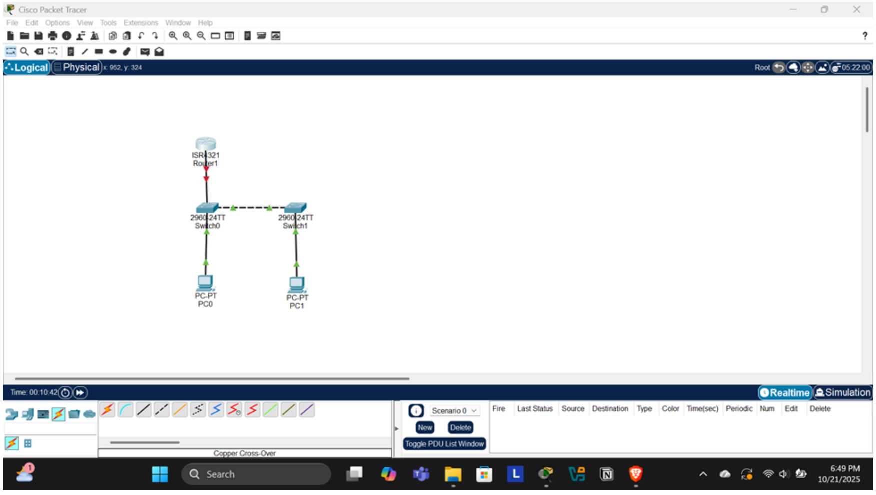
Part 2: Configure VLANs on switches

Part 3: Configure Switch Security

ScenarioPart 1: Configure the Network

Step 1: Cable the network

The network was configured using the topology highlighted below:

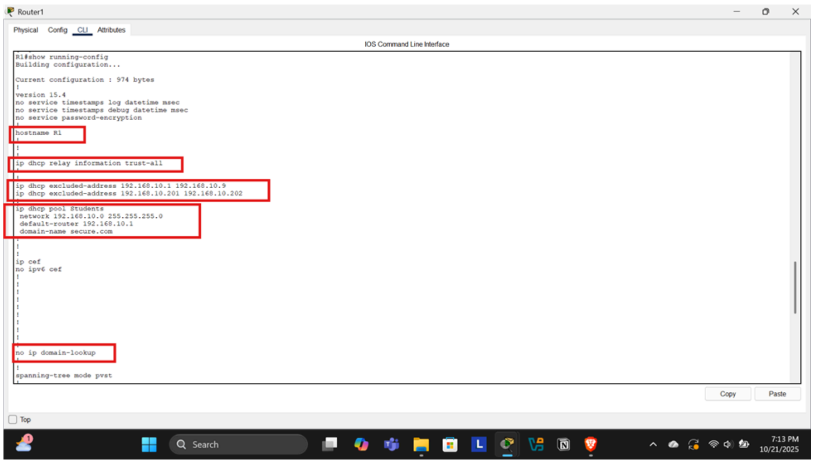


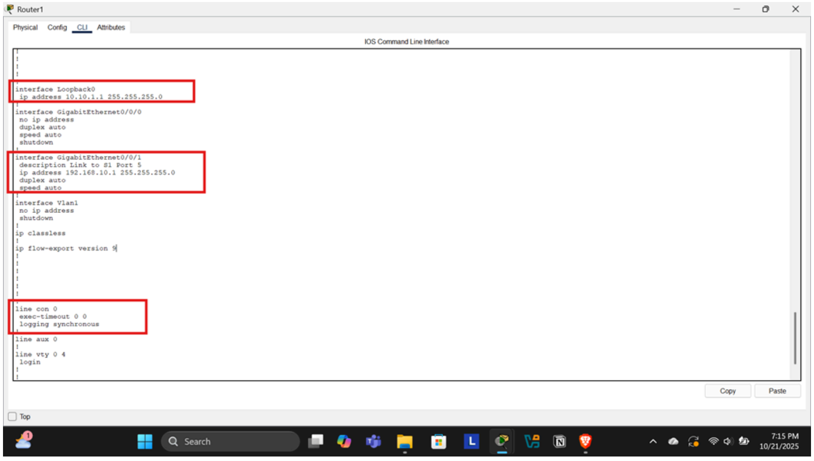
Required resources:

- 1 Router Cisco 4321
- 2 Switches Cisco 2960
- 2 PCs
- Copper Straight-through cable to connect different devices
- Copper Cross Over cables to connect similar devices

Step 2: Configure R1

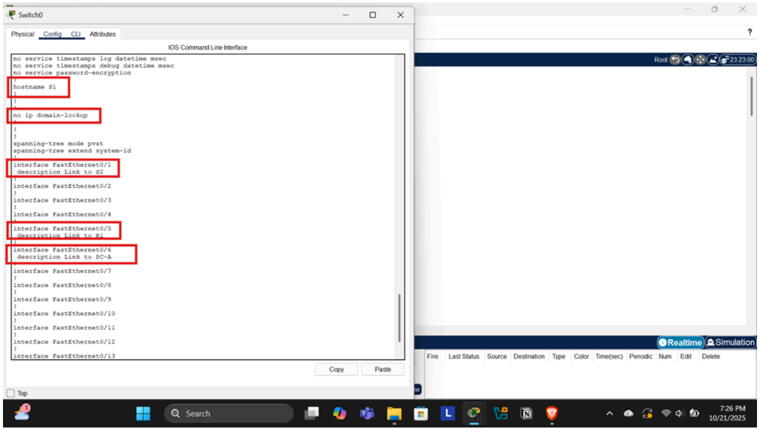
1. Loaded the configuration script on R1 and verified using show running-config as highlighted below.



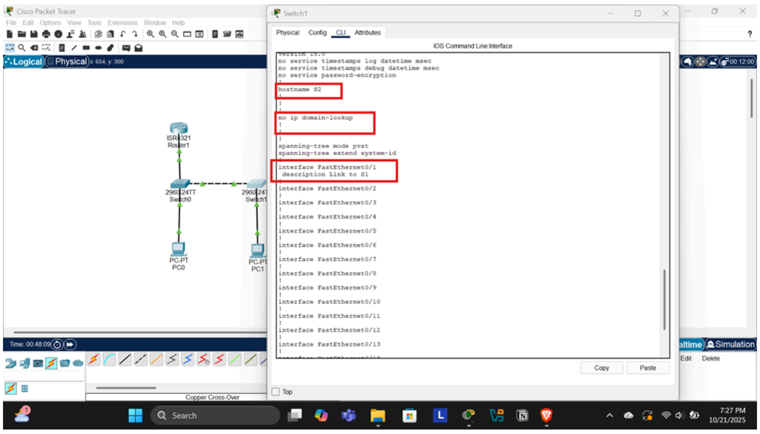


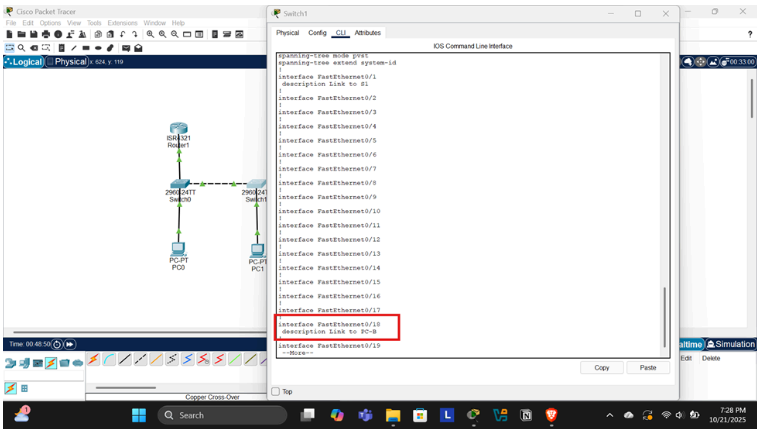
Step 3: Configure and verify basic switch settings

1. Configured the hostnames, prevented unwanted DNS lookups on both switches, configured the interfaces with their descriptions, and the default gateway on both switches
    a. Switch 1



    b. Switch 2

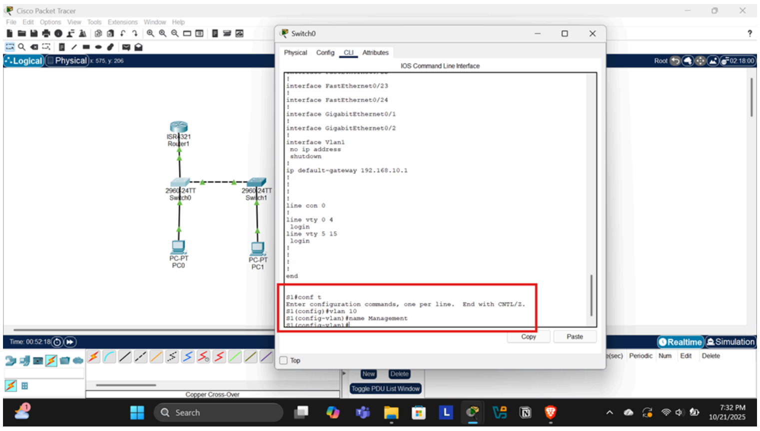




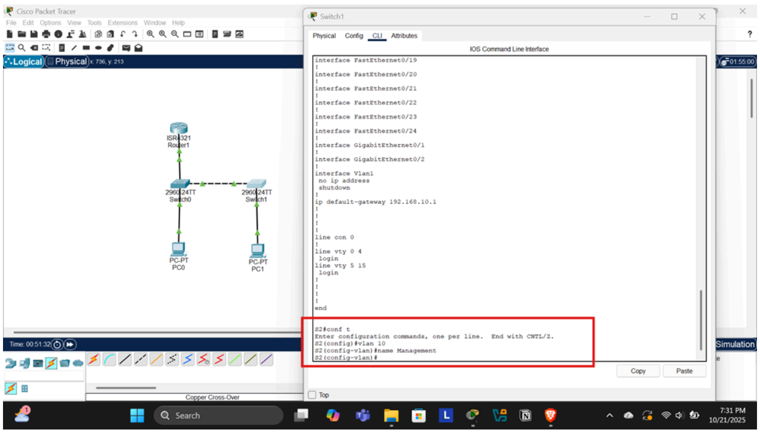
Part 2: Configure VLANs on Switches

Step 1: Configure VLAN 10

1. Configure VLAN 10 to S1 and S2
    a. Switch 1

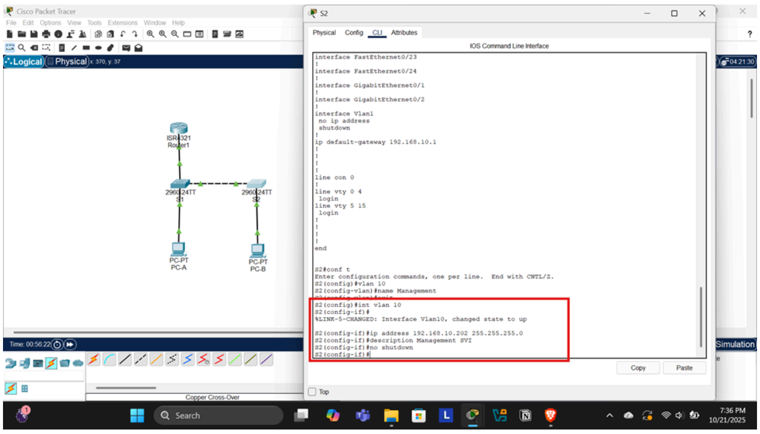


    b. Switch 2

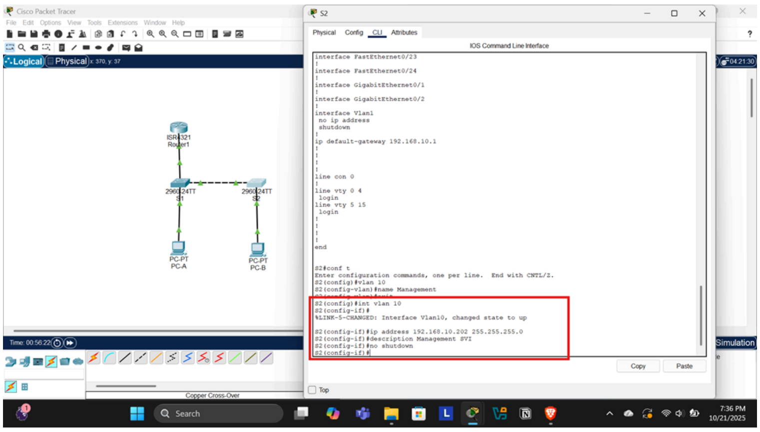


Step 2: Configure the SVI for VLAN 10

1. Configure the IP addresses for VLAN 10 on both switch 1 and 2
    a. Switch 1

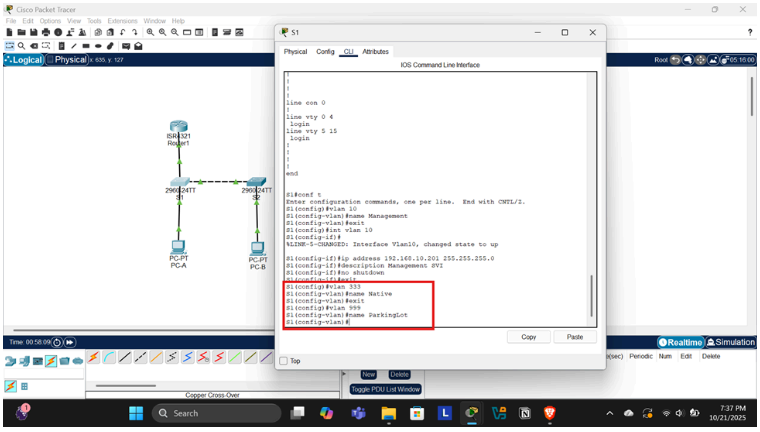


    b. Switch 2

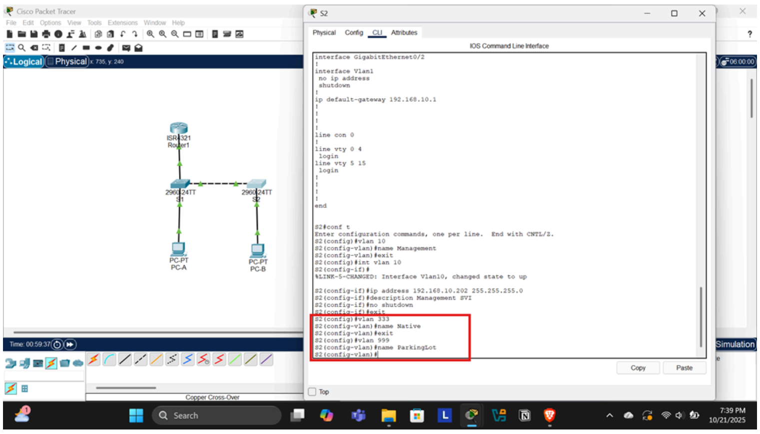


Step 3: Configure VLAN 333 with the name native on S1 and S2

1. Configure VLAN 333 with the name Native, and VLAN 999 on switches 1 and 2.
    a. Switch 1



b. Switch 2

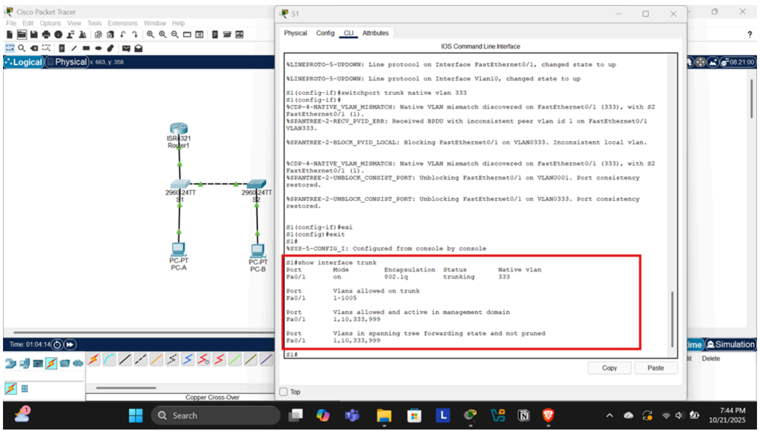


Part 3: Configure Switch Security

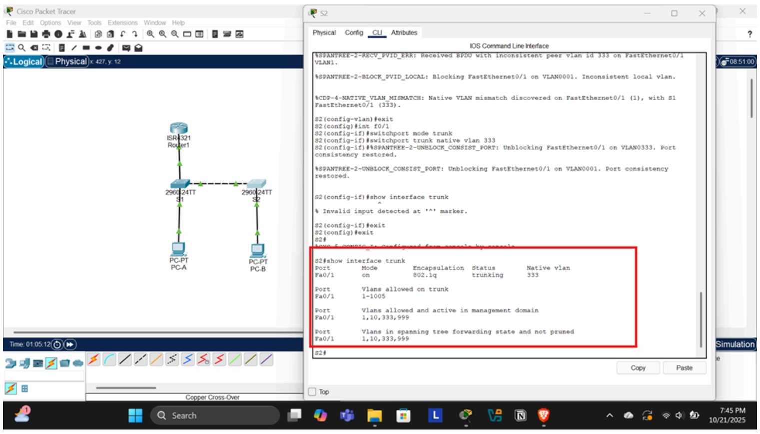
Step 1: Implement 802.1Q Trunking

1. Configure trunking on F0/1 to use VLAN 333 as the native VLAN and verify on both switches

   a. Switch 1

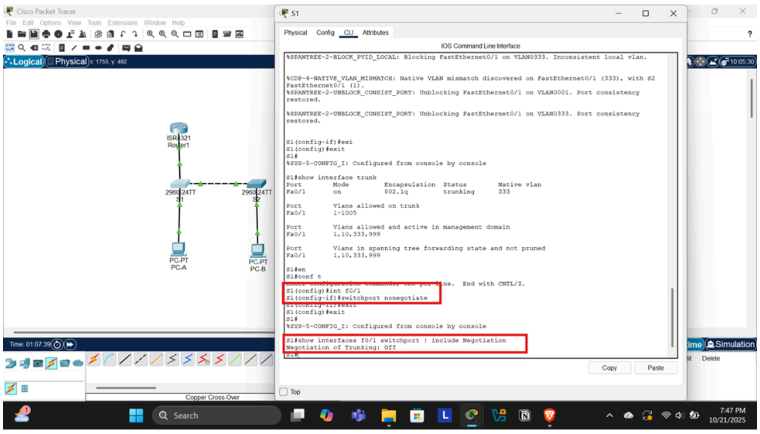


b. Switch 2

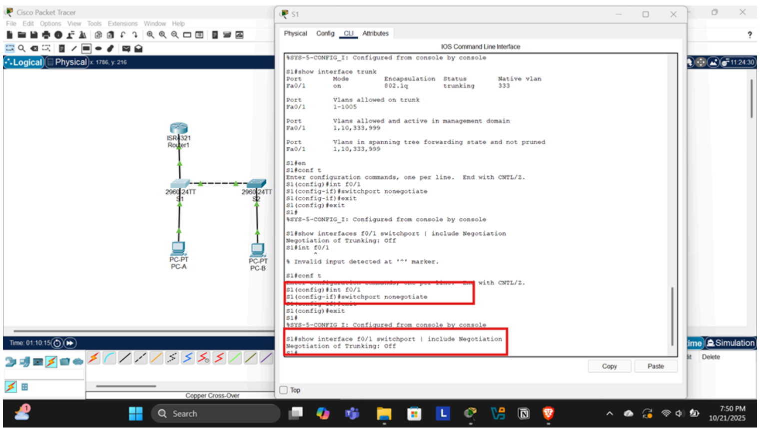


2. Disable DTP Negotiations on both switches and verify

a. Switch 1

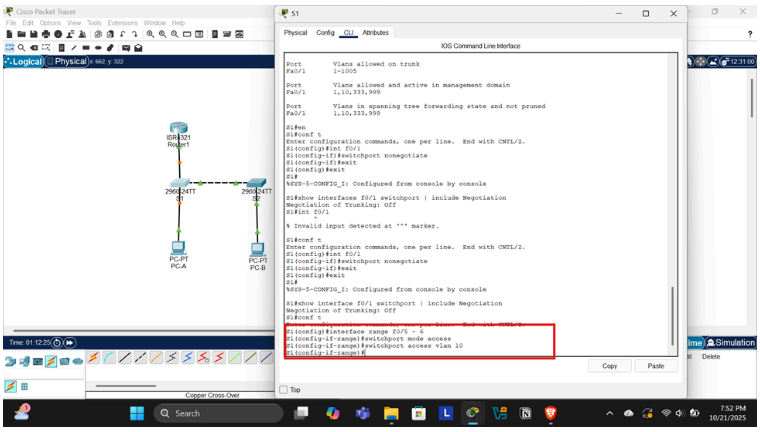


b. Switch 2

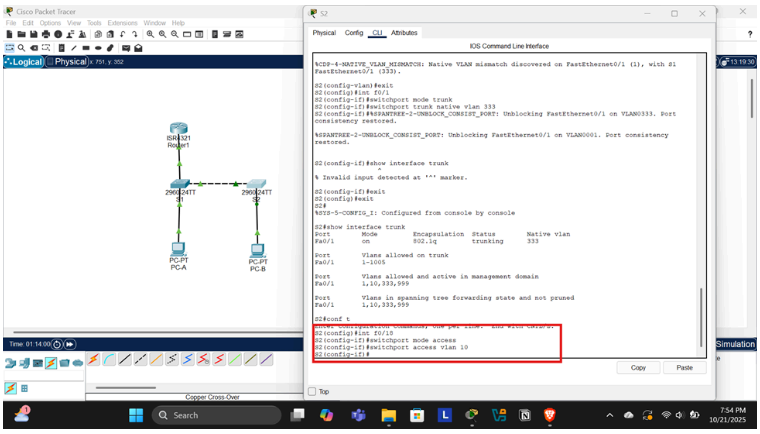


Step 2: Configure Access ports

1. Configure Access ports that are associated with VLAN 10 on both switches
    a. Switch 1

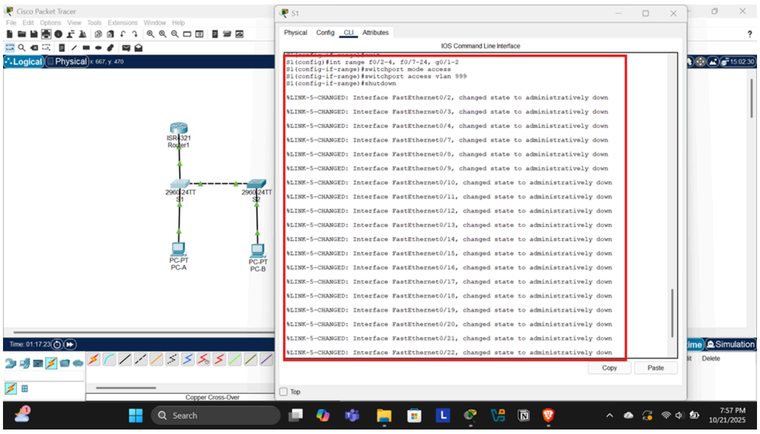


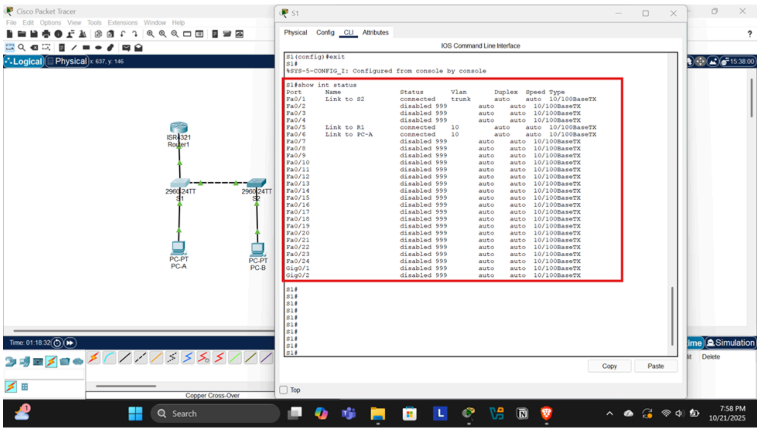
    b. Switch 2



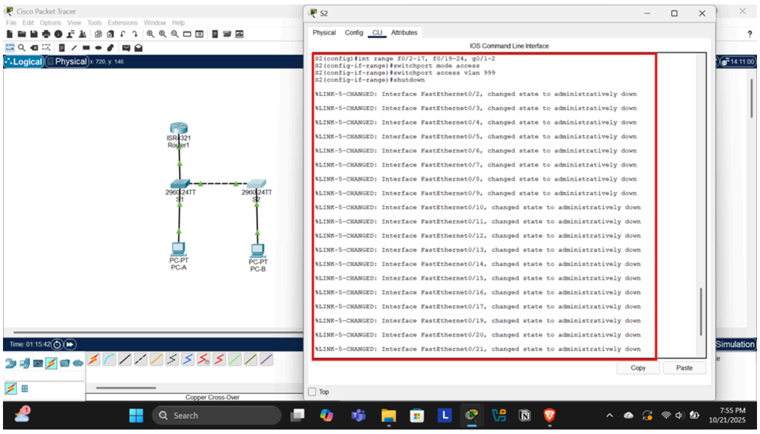
Step 3: Secure and disable unused switchports

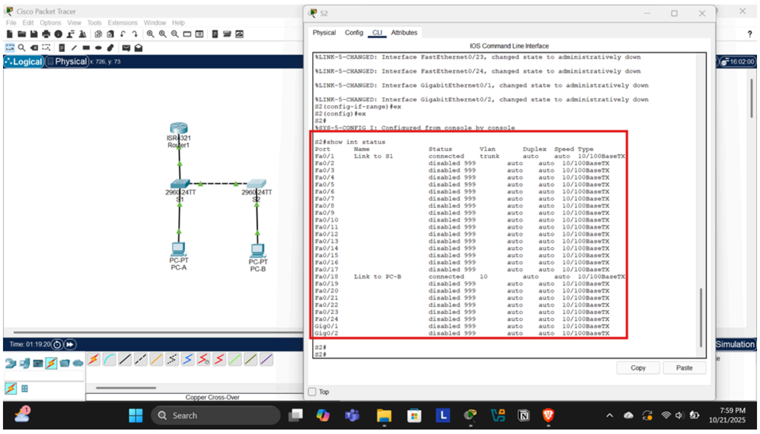
1. Move the unused ports to VLAN 999, disable them, and verify
    a. Switch 1





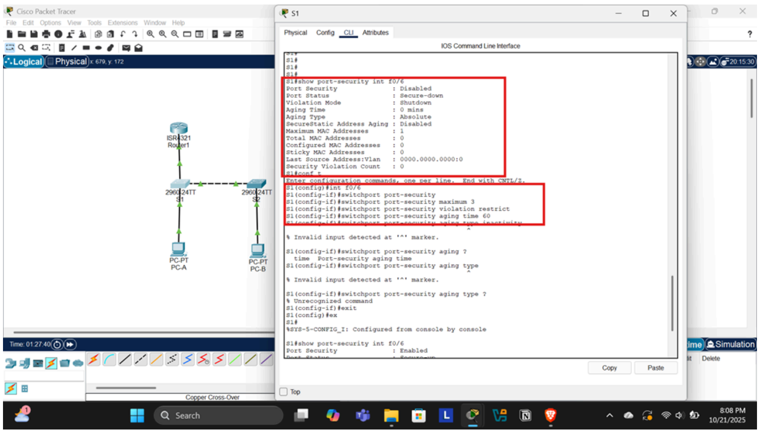
b. Switch 2

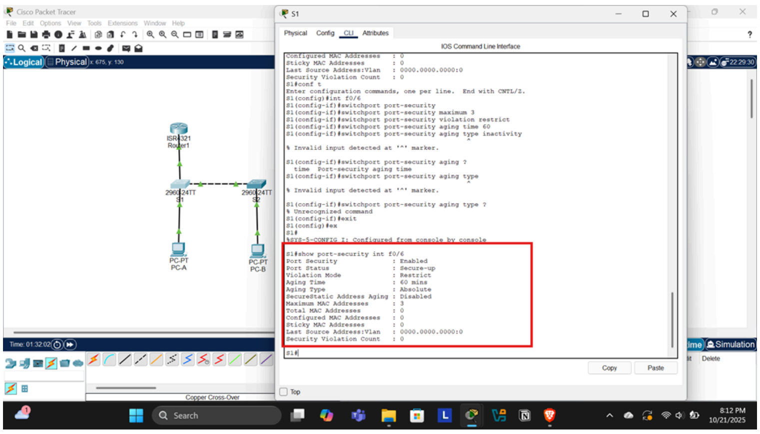




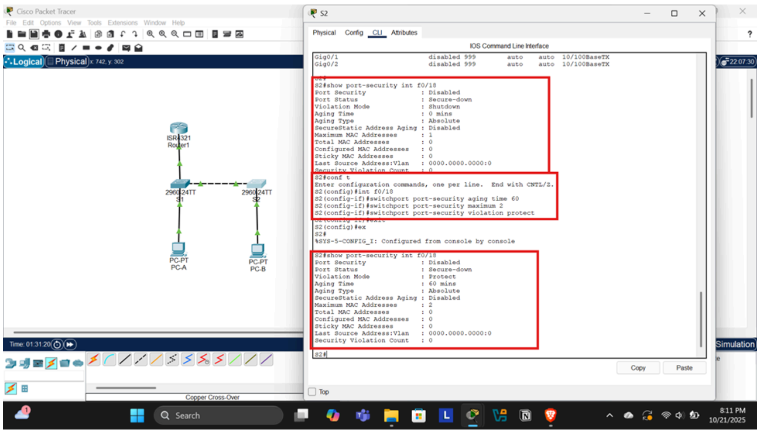
Step 4: Document and implement port security features

1. Configured port security on the access ports on both switches
    a. Switch 1



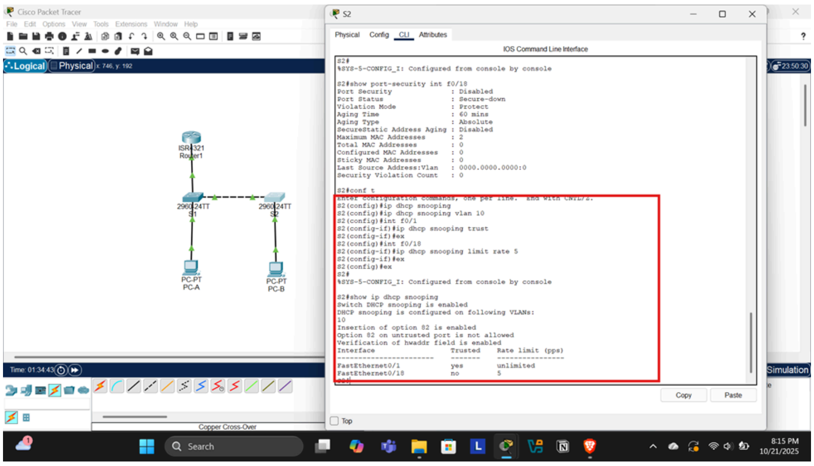


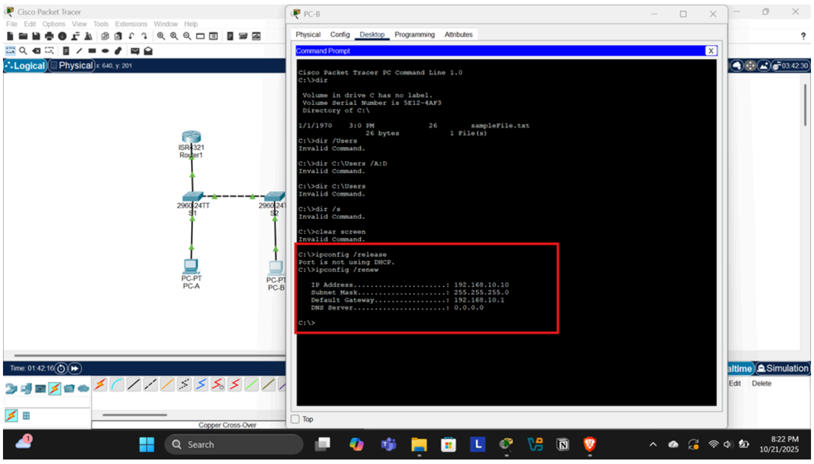
    b. Switch 2

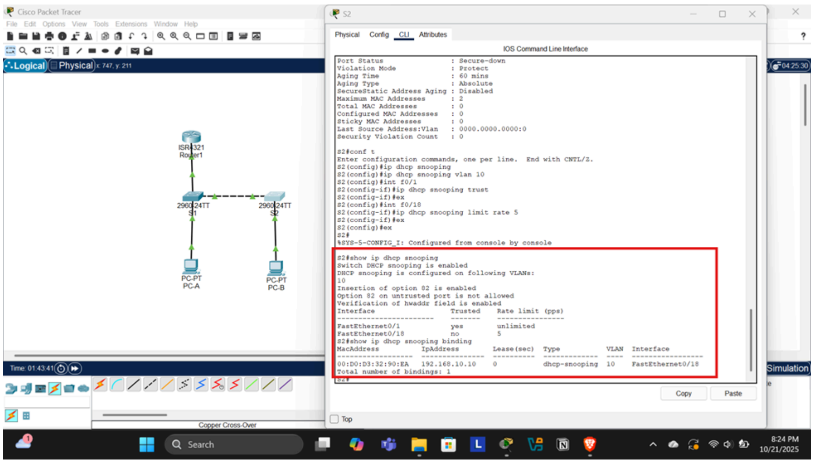


Step 5: Implement DHCP snooping security

1. Enabled DHCP snooping and configured DHCP snooping on VLAN 10, then verified.

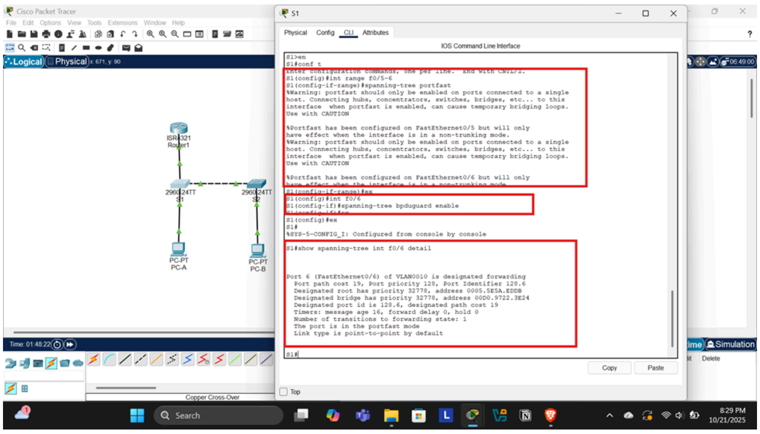




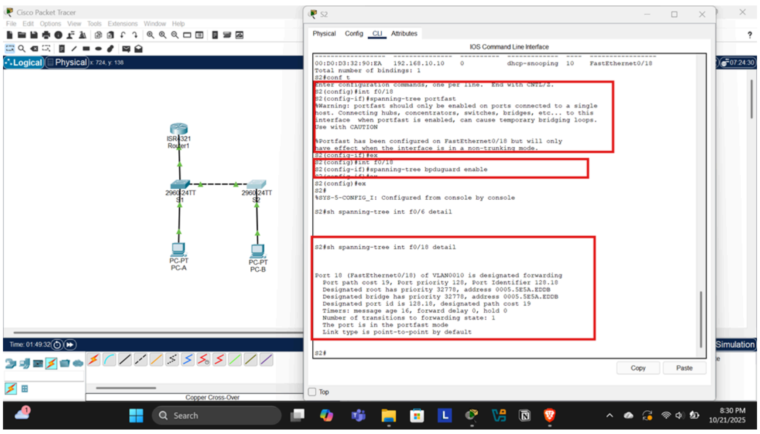


Step 6: Implement PortFast and BPDU guard.

1. Configured PortFast and enabled BPDU on the access ports on both switches
    a. Switch 1

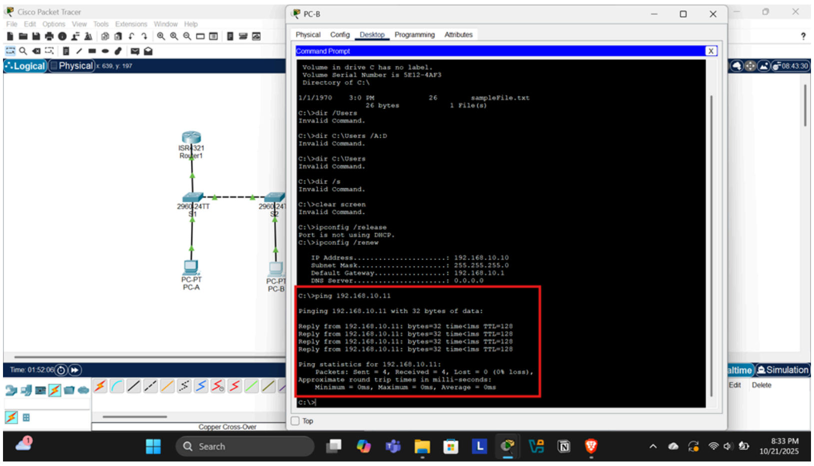


    b. Switch 2

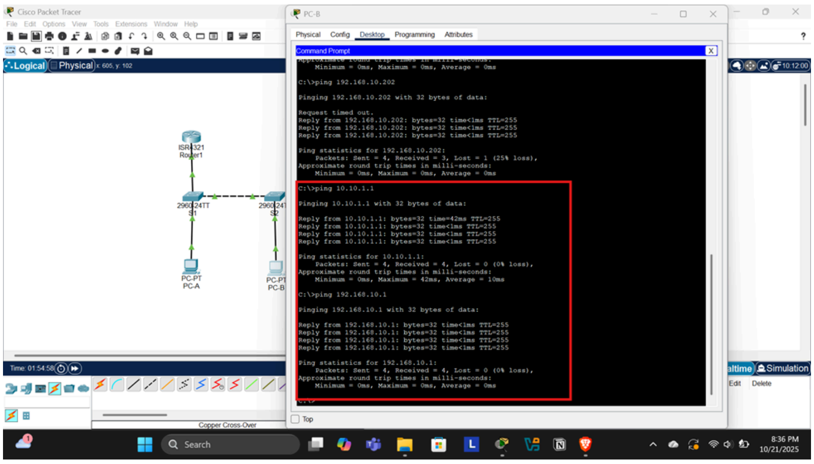


Step 7: Verify end-to-end connectivity

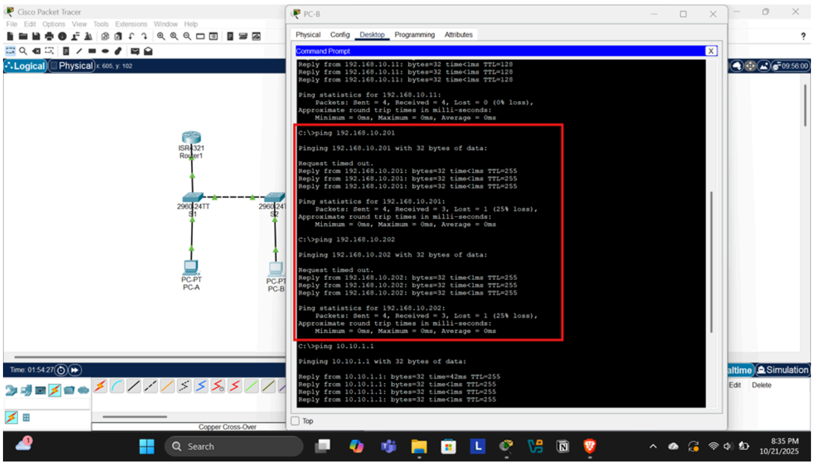
1. PC B to PC A



2. PC B to loopback on R1 and R1



3. PC B to S1 and S2



Conclusion

In conclusion, the network configuration process was successfully completed by setting up VLANs, securing unused ports, and implementing port security features across both switches. Additionally, DHCP snooping and BPDU guard were employed to enhance network security. End-to-end connectivity was verified, ensuring effective communication between all devices. This implementation lays a solid foundation for a secure and efficient network environment.